

# Week 9: Intro to Bayes

09/03/26

## Happiness

### Question 1

Consider the happiness example from the lecture, with 118 out of 129 women indicating they are happy. We are interested in estimating  $\theta$ , which is the (true) proportion of women who are happy. Calculate the MLE estimate  $\hat{\theta}$  and 95% confidence interval.

```
y <- 118
```

```
n <- 129
```

```
y/n
```

```
[1] 0.9147287
```

```
p <- y/n
```

```
se <- sqrt(p* (1-p)/n)
```

```
p - 1.96*se
```

```
[1] 0.8665329
```

```
p + 1.96*se
```

```
[1] 0.9629244
```

## Question 2

Assume a Beta(1,1) prior on  $\theta$ . Calculate the posterior mean for  $\hat{\theta}$  and 95% credible interval.

```
a <- b <- 1  
  
p_bayes <- (y+a)/(n+a+b)  
p_bayes
```

```
[1] 0.9083969
```

```
qbeta (0.025, y+a, n-y+b)
```

```
[1] 0.8536434
```

```
qbeta (0.925, y+a, n-y+b)
```

```
[1] 0.9420615
```

## Question 3

Now assume a Beta(10,10) prior on  $\theta$ . What is the interpretation of this prior? Are we assuming we know more, less or the same amount of information as the prior used in Question 2? we can also put a b in formula, instead of 10.

```
a <- b <- 10  
  
p_bayes <- (y+a)/(n+a+b)  
p_bayes
```

```
[1] 0.8590604
```

```
qbeta (0.025, y+a, n-y+b)
```

```
[1] 0.7990363
```

```
qbeta (0.925, y+a, n-y+b)
```

```
[1] 0.8981297
```

```
library(tidyverse)
dp <- read_csv("/Users/liangqishen/Desktop/data/wpp_pop_clean.csv")
df <- read_csv("/Users/liangqishen/Desktop/data/WPP2024_Demographic_Indicators_Medium.csv")
dpf <- dp |> filter(region=="Canada", sex == "Female") |>
  pivot_longer(x0:x100) |>
  mutate(age = as.numeric(str_remove(name, "x")))

dpm <- dp |> filter(region=="Canada", sex == "Male") |>
  pivot_longer(x0:x100) |>
  mutate(age = as.numeric(str_remove(name, "x")))

breaks <- c(seq(0,100, by = 5), Inf)
labels <- seq(0,100, by = 5)

# Create the age groups
dpf$age_group <- cut(dpf$age,
  breaks = breaks,
  labels = labels,
  right = FALSE)

dpf5 <- dpf |>
  group_by(year, age_group) |>
  summarize(pop = sum(value))

dpm$age_group <- cut(dpm$age,
  breaks = breaks,
  labels = labels,
  right = FALSE)

dpm5 <- dpm |>
  group_by(year, age_group) |>
  summarize(pop = sum(value))

dp5_23 <- dpf5 |> filter(year==2023)

C <- dp5_23$pop[dp5_23$age_group==0] + dpm5$pop[dpm5$age_group==0&dpm5$year==2023]
W <- dp5_23$pop[dp5_23$age_group==15]+dp5_23$pop[dp5_23$age_group==20]+dp5_23$pop[dp5_23$age_group==25]
```

```
p <- 1/7
s <- 1

iTFR <- 1/s*1/p*C/W
iTFR
```

```
[1] 1.483144
```

```
x_multiplier <- (10.65-12.55*(dp5_23$pop[dp5_23$age_group==25]+dp5_23$pop[dp5_23$age_group==25]))
xTFR <- x_multiplier*C/W
xTFR
```

```
[1] 1.44793
```

```
# the actual TFR is
df |> filter(Location=="Canada") |> select(Time, TFR)
```

```
# A tibble: 152 x 2
   Time    TFR
  <dbl> <dbl>
1  1950  3.37
2  1951  3.42
3  1952  3.56
4  1953  3.64
5  1954  3.74
6  1955  3.75
7  1956  3.77
8  1957  3.85
9  1958  3.81
10 1959  3.86
# i 142 more rows
```